

What Keeps a Planet in Orbit?

PhET Gravity and Orbits · PhET

Year 10

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/gravity-and-orbits>

Curriculum links: AC9S10U05 · AC9S7U04 · AC9S7U03

Learning intentions

- I can explain how gravity acts as the force that keeps objects in orbit.
- I can predict what happens to an orbit when I change an object's mass or starting velocity.
- I can relate the force, mass and motion of orbiting bodies using Newton's laws.

Getting started

Open PhET Gravity and Orbits and start on a screen with a star and planet (or Earth and a satellite). Make sure gravity is switched on. Watch a full orbit before changing anything, then use the tasks below to test what affects the shape and stability of the orbit.

Tasks

1. With gravity on and default settings, describe the shape of the orbit you see. Is it a perfect circle?

2. Switch gravity off while the object is moving. What happens to its path? Explain this using Newton's first law (an object in motion stays in motion unless acted on by a force).

3. Turn gravity back on and increase the mass of the central body (e.g. the star). What happens to the orbiting object's path or speed?

4. Test at least four different starting velocities for the orbiting object and record what happens to its orbit each time.

Starting velocity	What happens to the orbit

5. Based on your table, explain what range of starting velocities keeps an object in a stable orbit, and what happens outside that range.

6. Increase the mass of the orbiting object (not the central body) while keeping everything else the same. Does the shape of its orbit change? Is this what you expected?

7. Use Newton's laws to explain why a more massive central body would require a planet to orbit faster to stay in a stable orbit at the same distance.

8. The Moon orbits Earth roughly every 27 days, contributing to predictable tides. Based on what you've observed in this simulation, what would happen to the Moon's orbit if Earth's mass suddenly increased?

Extension (optional)

Satellites in 'geostationary' orbit stay above the same point on Earth at all times. Using what you've learned about the relationship between orbital distance, speed and gravity, explain what would need to be true about a satellite's velocity for this to work.
